

[This question paper contains 4 printed pages.]

Your Roll No.....

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Sr. No. of Question Paper : 13202

Unique Paper Code : 2344000006

Name of the Paper : Advanced Web Programming

Name of the Course : **Generic Elective - Computer Science Programme**

Semester : V

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. **Section A** is compulsory.
3. Attempt any **Four** Questions From **Section B**.
4. Part of a question must be answered together.

Section A

(Compulsory)

1. (a) Explain the use of `$_REQUEST`. How is it different from `$_GET` and `$_POST`? (3)
- (b) How do you write a comment in PHP? Give examples of any two types. (3)
- (c) What is URL encoding? Why is it necessary when sending form data using the GET method? (3)
- (d) What is the role of MySQL in Server-side Programming using PHP. (3)
- (e) What is the difference between `array_merge()` and `array_combine()` functions in PHP? Provide an example for each. (3)
- (f) Write the output of the following PHP code: (3)

```
<?php
$a = "12";
$b = 6;
$c = $a + $b;
$d = $a . $b;
echo $c . " - " . $d;
?>
```
- (g) What is jQuery's `$(document).ready()` function used for? Give one example. (3)
- (h) What is the purpose of preprocessing form inputs before validation? Discuss any two techniques used for preprocessing user input. (3)

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- (i) What will be the output of the following code? Justify your answer. (3)
- ```
<?php
$p = 8;
$q = $p;
$q = $q + 4;
unset($p);
echo $q;
?>
```
- Handwritten notes:*  $q = 8$   
 $q = 8 + 4 = 12$   
~~unset(\$p)~~
- (j) Write a PHP script that prints "Excellent" if marks is 90 or above, "Good" if marks is 75 or above, "Average" if marks is 50 or above, otherwise "Fail". (3)

### Section B

(Attempt any 4 Questions)

(Parts of the Questions must be answered together)

2. (a) Write an HTML snippet that creates a textbox and a button labelled "Reverse Text". After clicking the button, the reverse text of the string entered in textbox should be displayed. (5)
- (b) Differentiate between: (5)
- `==` and `===`
  - `include` and `require`
- Explain with the help of suitable examples of each.
- (c) What is an associative array? Write a PHP script to create an associative array storing 5 countries and their capitals. (5)
3. (a) Write a PHP function `findPrice($item)` that searches through a list of product names and a matching list of prices. The function should return the price of the product whose name matches the given item. (5)
- (b) Create an HTML form titled 'Student Feedback Form' with fields: (5)
- Student Name (text)
  - Course (text)
  - Rating (radio buttons: 1–5)
  - Suggestions (textarea)
  - Submit button
- (c) Write a PHP script to Check if Student Name and Rating are provided in part (b). If missing, display appropriate error messages. Otherwise, print all the submitted details on a new webpage named "Welcome.php". (5)



4. (a) Write a PHP script which counts the number of times the word "web" (case-insensitive) appears in the following string: "Web technologies form the backbone of modern WEB applications." (5)
- (b) Write a jQuery code using the `click()` event to hide all `<div>` elements when a button with id `btnHide` is clicked. (5)
- (c) Write a PHP function `replaceWord($str)` that takes a string containing multiple words and replace the longest word with the smallest word in the string and returns the modified string. (5)
5. (a) Use: (5+5)
- Host: localhost
  - DB: CollegeDB
  - User: root
  - Password: pass123
- i. Write PHP code to connect to the database and display "Connection Successful" if no error occurs; otherwise terminate with the error message.
- ii. Using MySQL, write PHP scripts to:
- Insert a student whose ID=20, Name="Meera", Dept="IT", and Score=92
  - Update Score of student with 88 whose ID=5
  - Delete record of the student whose Name is "John"
  - Close the database connection.
- (b) Create an HTML form that asks the user to enter a paragraph. Write a PHP script that: (5)
- i. Split the paragraph entered into sentences
  - ii. Counts number of sentences containing more than 10 words
  - iii. Find the sentences with the maximum number of words
6. (a) How does PHP perform type casting? Explain with an example. (5)
- (b) Using jQuery, write a script that: (5)
- i. Hides all paragraphs when a button is clicked.
  - ii. Shows them again with a slow effect.
  - iii. Changes their text color to blue on mouseover.
  - iv. Adds a CSS class named "highlight" to them.
  - v. Removes the class "highlight" added above when the mouse leaves.

- (c) Write a code snippet using JSON that performs the following tasks: (5)
- Create a JSON object containing details of a student (name, age, course, marks).
  - Display all the values stored in the JSON object.
  - Convert the JSON object into a JSON string using `JSON.stringify()` and print it.
  - Convert the JSON string back into an object using `JSON.parse()`.

- 7x (a) Write a PHP program to create an array A of integers and perform the following tasks: (5)
- Remove all duplicate values from array A.
  - Sort the array A in descending order.
  - Find the second largest number in the array A.
  - Display the final array A.

- (b) Find all errors and rewrite the corrected code. (5)

```
<?php
funct addNumbers($a, $b)
{
 total = $a + $b
 return total;
}

$num1 == 15;
$num2 = "20;

$result = addNumbers($num1 $num2);
echo "Sum is : " $result "
"

if($result % 2 = 0)
{
 echo "Number is Even"
}
else
{
 echo "Number is Odd";
?>
```

- (c) What are jQuery Effects and Animations? Describe their purpose and explain in detail how the methods `fadeIn()` and `fadeOut()` work. Provide suitable examples for each. (5)